



Embracing Momentary Focus for Elevated Happiness: The Moderating Effects of External Motivation and Trait Mindfulness

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Abstract

Objectives The ability to focus on the present moment, also known as momentary focus, is recognized as a crucial psychological mechanism associated with state mindfulness and flow experience, potentially serving as a vital pathway to increasing current happiness. Motivation, an intrinsic or extrinsic drive to achieve goals, can influence our emotional state during momentary focus. However, awareness of motivation may be affected by individual differences in trait mindfulness, determining whether we are influenced by them. This study explored the moderating effects of motivation on momentary focus to predict happiness and examined whether individuals with high trait mindfulness were especially susceptible to the influence of motivations.

Method We used the experience sampling method to capture the daily experiences of 56 Taiwanese college students over a 2-week span. Participants were required to report their current momentary focus, happiness, and motivation for engaging in activities three times a day.

Results The findings demonstrated that momentary focus positively predicted happiness. However, higher external motivation weakened the positive association between momentary focus and happiness. Furthermore, individuals with high trait mindfulness were more sensitive to external motivation, which amplified the moderating effects of external motivation on the positive relationship between momentary focus and happiness.

Conclusions These findings highlight the importance of minimizing the influence of external motivation in mindfulness training, particularly for individuals with high trait mindfulness.

Preregistration This study was not preregistered.

Keywords Momentary focus · Happiness · Motivation · Mindfulness · Flow theory

Momentary focus refers to a state of heightened attention to the present moment. A lack of momentary focus has become a common phenomenon in modern life, negatively impacting individuals' happiness. Previous research found that, during 46.9% of the sampled time, people were unable to focus on their current activities, and this mind-wandering state was associated with reduced happiness (Killingsworth & Gilbert, 2010). Such unintentional distraction can also impair performance on tasks such as reading comprehension, working memory, and intelligence tests (Mooneyham & Schooler,

2013). Moreover, drivers whose minds wander tend to drive faster, experience longer reaction times in unexpected situations, and are more prone to causing accidents (Yanko & Spalek, 2014). Given the detrimental effects of being unfocused, this study explores the mechanisms through which momentary focus contributes to happiness.

Momentary focus lies at the intersection of state mindfulness and flow experience. This shared characteristic of state mindfulness and flow experience may be a key factor in fostering happiness. State mindfulness, characterized by sustained attention to both internal and external stimuli, leads to a deeper understanding of one's thoughts and feelings (Dreyfus, 2011). Research suggests a strong link between state mindfulness and increased happiness (Ding et al., 2019; Rowland et al., 2020; Ryan & Deci, 2000). Flow experience, on the other hand, involves complete immersion in an activity, often leading to a loss of self-awareness and a distorted

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sense of time (Csikszentmihalyi, 1990). Studies have shown that flow experience positively correlates with happiness (Asakawa, 2010; Bryce & Haworth, 2002; Collins et al., 2009). While state mindfulness is related to internal experiences and flow experience is related to external engagement, both states require a high degree of momentary focus. This shared characteristic of momentary focus may explain the positive correlation between state mindfulness and flow experience (Schutte & Malouff, 2023). Given that momentary focus appears to be a core mechanism shared by state mindfulness and flow experience, and both can positively predict happiness, this study hypothesizes that momentary focus can positively predict happiness.

Existing studies have demonstrated that momentary focus offers numerous benefits, notably enhancing positive emotions in daily life (Blanke et al., 2018; Felsman et al., 2017; Rush & Grouzet, 2012). Momentary focus is a process of being present, but some researchers acknowledge that an excessive emphasis on being present may also yield certain disadvantages. Research has indicated that a present-time perspective can lead to heightened impulsivity and aggression, which, in turn, may contribute to risky driving behavior (Zimbardo et al., 1997). Furthermore, individuals who adopt a present-oriented time perspective when considering significant life goals are more susceptible to engaging in substance abuse, thereby fostering ill-being (Keough et al., 1999). Given the mixed results regarding the relationship between momentary focus and happiness, exploring potential moderating factors that delineate when momentary focus engenders either heightened or diminished happiness becomes valuable.

Motivation, an intrinsic or extrinsic drive to achieve goals, can influence our emotional state during momentary focus. Ryan and Deci (2000) formulated the self-determination theory, classifying trait motivation into three primary types: intrinsic motivation, extrinsic motivation, and amotivation. Guay et al. (2000) extended the core framework of self-determination theory, suggesting that extrinsic motivation can be divided into two types of situational motivation: identified motivation and external motivation, with the goal of better capturing situational motivation during activities. Intrinsic motivation stems from inherent satisfaction and the desire for exploration, while identified motivation manifests when individuals engage in activities they value, perceive as important, and choose autonomously. In addition, external motivation pertains to behaviors driven by external rewards or punishments. Conversely, amotivation characterizes individuals who act without intention or fail to engage in any action (Ryan & Deci, 2017). Both intrinsic motivation and identified motivation are considered forms of autonomous motivation, while external motivation and amotivation are regarded as forms of controlled motivation (Ryan & Deci, 2017).

Autonomous motivation and controlled motivation, respectively, enhance and diminish the relationship between momentary focus and happiness. Previous meta-analysis studies have found a positive correlation between intrinsic motivation, identified motivation, and trait mindfulness (Donald et al., 2020; Li et al., 2023). Donald et al. (2020) reported a negative correlation between external motivation, amotivation, and trait mindfulness. Ryan et al. (2021) found that trait mindfulness enhances present-moment awareness, promoting the integration of self-values and ultimately leading to higher autonomous motivation and lower controlled motivation. Nakamura and Dubin (2015) found a positive relationship between intrinsic motivation and flow experience, and Rheinberg (2020) suggested that intrinsic motivation serves as an incentive during activities, facilitating the flow experience. In addition, Csikszentmihalyi (1990) posited that flow experience—characterized by fully enjoying the present moment—occurs only when individuals disregard external outcomes.

The above-mentioned studies on state mindfulness and flow experience indicate that autonomous motivation (intrinsic and identified motivation) promotes momentary focus, while controlled motivation (external motivation and amotivation) diminishes momentary focus. However, past studies have primarily focused on trait motivation, with limited exploration of how situational motivation in specific activities influences the positive relationship between momentary focus and happiness. This study hypothesizes that when individuals engage in activities driven by higher intrinsic or identified motivation, the positive relationship between momentary focus and happiness will be strengthened. Conversely, when individuals engage in activities driven by external motivation or amotivation, the link between momentary focus and happiness will be weakened.

Individuals with high trait mindfulness may be particularly sensitive to motivation, thereby enhancing its moderation effect on the relationship between momentary focus and happiness. Ryan et al. (2021) suggested that the trait mindfulness promotes a deeper awareness of internal phenomena, including desires, emotions, needs, conflicts, and stresses. Past research has also found that short sessions of mindfulness meditation increase individuals' awareness of their motivations (Smyth & Milyavskaya, 2021). Based on the above evidence, we infer that individuals with high trait mindfulness may be more aware of their motivations. Thus, the impact of motivation on focus and happiness is stronger. Conversely, individuals with low trait mindfulness, who may not be clearly aware of their activity motivations, are unaffected by their happiness via increases or decreases in motivation.

The central objective is to verify a hypothesis that momentary focus is a positive predictor of enhanced happiness (Hypothesis 1). Moreover, intrinsic motivation

and identified motivation may enhance the positive relationship between momentary focus and happiness, while external motivation and amotivation hinder this relationship (Hypothesis 2). Additionally, individuals with higher trait mindfulness are more sensitive to the awareness of motivations, meaning their motivations have a stronger moderating effect on the relationship between momentary focus and happiness (Hypothesis 3).

Since this study aimed to measure state variables and capture the authentic nature of their experiences, the experience sampling method (ESM) is the most suitable approach. ESM is a data collection method where participants respond to repeated assessments at different points in time while operating in their natural settings (Scollon et al., 2003). Scollon et al. (2003) found that ESM offers high ecological validity compared to the artificial settings of laboratory experiments. Additionally, this method minimizes memory bias, enabling the accurate capture of individuals' psychological states at the moment. Furthermore, state variables fluctuate with ongoing activities, and ESM is well-suited to track the dynamic within-person processes involving variability over time (Fisher & To, 2012).

Method

Participants

Sixty-three Taiwanese participants were recruited for this study through web postings or advertisements in introductory psychology courses. According to Arend and Schäfer (2019), to achieve a power of 0.80 or higher, at least 30 participants (Level 1) are needed, and a total of at least 200 data entries (Level 2) are required for multilevel regression analysis. The sample size of this study meets these standards.

As an incentive for participation, individuals could opt for either 180 NT dollars or one-course credit upon completing over 80% of the total survey. However, if the response rate fell below 80% but exceeded 50%, the monetary and credit rewards would be halved. There would be no monetary reward if the response rate was below 50%. Seven participants with a response rate of less than 20% were excluded from the study.

Ultimately, 56 participants (comprising 47 females and 9 males) were retained for data analysis. Their ages exhibited a mean of 19.95 years (range: 18–25; SD : 1.44). The participant count exceeded the minimum requirement of 50, which is necessary for conducting a multilevel analysis (Maas & Hox, 2005). All participants were duly informed and provided consent by completing the requisite informed consent forms.

Procedure

Participants completed a pre-test questionnaire encompassing demographic details and trait mindfulness before engaging in the experiment. Throughout the experiment, participants received daily reminders via the LINE communication software, prompting them to complete the questionnaire. They used the online questionnaire platform “Qualtrics” to respond to experience sampling items upon receiving the reminder. Participants were instructed to complete the questionnaire three times daily for 14 days, spanning the months of November to December 2022.

Reminders sent by LINE to fill in the questionnaire were scheduled with a minimum interval of 3 hr to ensure a comprehensive depiction of participants' states during various activities. Additionally, participants were only permitted to complete the questionnaire within 1 hr after receipt of the reminder. After the conclusion of the entire 14-day study, the experimenter disbursed payment to participants based on the frequency of their questionnaire completions. Among the 56 participants included in the analysis, response frequencies ranged from a minimum of 22 to a maximum of 42, yielding an overall response rate of 84.9%.

Measures

Trait Mindfulness Questionnaire

Baer et al. (2006) proposed that mindfulness can be defined as a *trait* variable and can be assessed across five facets: (a) Nonreactivity to inner experience, (b) Observing thoughts and feelings, (c) Acting with awareness, (d) Describing with words, and (e) Nonjudging of experience. Later, Huang et al. (2015) translated the Five Facet Mindfulness Questionnaire into Chinese and demonstrated that it exhibits good reliability and validity within the Taiwanese population. Examples of trait mindfulness items: *I can easily put my beliefs, opinions, and expectations into words; It seems I am “running on automatic” without much awareness of what I’m doing* (reverse item). A 6-point Likert scale was employed for 39 items, ranging from 1 (*Strongly disagree*) to 6 (*Strongly agree*). The internal consistency of the trait mindfulness measure was evaluated using Cronbach's alpha coefficient, yielding a value of 0.92, which indicates strong reliability (Cho, 2016). Participants, on average, reported moderate levels of trait mindfulness score ($M = 145.70$, $SD = 21.26$, range: 105–211).

Happiness and Momentary Focus Scale

Participants were instructed to provide real-time responses using a 6-point Likert scale to gauge their current happiness state, with 1 indicating *Very unhappy* and 6 indicating *Very*

happy. Subsequently, participants were required to assess their degree of momentary focus on a 6-point Likert scale, ranging from 1 (*Completely unfocused*) to 6 (*Completely focused*). The items of happiness and momentary focus are consistent with the items used in the experience sampling study conducted by Baranski et al. (2023). A similar item regarding momentary focus also appears in the Multidimensional State Mindfulness Questionnaire developed by Blanke and Brose (2017), the Chinese version of the Mindful Attention Awareness Scale (CMAAS) developed by Chang et al. (2011), and the Flow State Scale developed by Peng and Shen (2016). The Mindful Attention Awareness Scale and Flow State Scale demonstrate good reliability and validity within the Taiwanese population (Chang et al., 2011; Peng & Shen, 2016). According to an established standard (Chiu et al., 2015), ICC values exceeding the cutoff points of 0.06 for ICC(1) and 0.70 for ICC(2) are considered reliable for situational variables at the individual level. The ICC(1) of happiness in this study is 0.26, and the ICC(2) is 0.92. The ICC(1) of momentary focus in this study is 0.21, and the ICC(2) is 0.90. These results indicate that the items for happiness and momentary focus demonstrate good reliability in this study.

Situational Motivation Scale

After completing the items on momentary happiness and momentary focus, participants were prompted to identify their ongoing activities from a set of 23 options. The 23 activity types adopted and revised the work categories mentioned by Killingsworth and Gilbert (2010) in their experience sampling study. Based on the results of a pilot study, their “home computer” category was removed, and two new categories, “in class” and “prepare homework,” were added to align more closely with the lifestyles of Taiwanese people. Furthermore, participants were requested to elucidate intrinsic motivation, identified motivation, external motivation, and amotivation for engaging in the current activity, with the motivation items drawn from the Situational Motivation Scale (SIMS) (Guay et al., 2000).

To streamline the questionnaire for the experience sampling method, the original 16 SIMS questions were condensed to four items, each representing a distinct type of motivation. Specifically, intrinsic motivation was captured by the item “Because I think that this activity is interesting,” identified motivation by “Because I think that this activity is good for me,” external motivation by “Because I feel that I have to do it,” and amotivation by “I am doing this activity, but I am not sure if it is worth it.”

Importantly, all motivation items in our study surpassed the reliability thresholds, as indicated by the following ICC values: intrinsic motivation (ICC(1): 0.17, ICC(2): 0.88), identified motivation (ICC(1): 0.18, ICC(2): 0.89), external

motivation (ICC(1): 0.14, ICC(2): 0.85), and amotivation (ICC(1): 0.30, ICC(2): 0.94).

Data Analyses

When examining the main effect of momentary focus on happiness and the moderating effect of external motivation, the analytical approach used in this study aligns with the experience sampling studies conducted by Blanke et al. (2018) and Felsman et al. (2017). Specifically, we adopted multilevel regression analysis to predict both main effects and interactions. All occasional factors (including momentary focus, happiness, and motivation in three times a day for 14 days) were centered around individuals’ personal means. Subsequently, the person mean of each occasional factor was included, with the main effects and interactions set as both fixed and random effects. This approach allowed us to control for variations in intercepts and slopes across participants and to examine within-subject main effects and interactions.

When assessing the cross-level moderating effect of trait mindfulness on momentary focus, happiness, and external motivation, we adopted an analytical strategy consistent with Zhang et al. (2020). Multilevel regression was used to predict cross-level interactions. First, the trait mindfulness was centered around the grand mean to ensure the independence between independent and moderating variables. Additionally, since the trait mindfulness operates at the person level, it was subjected to person-mean centering and treated as a fixed effect. All occasional factors were included as both fixed and random effects during the analysis.

The multilevel regression analyses in this study were conducted using the “lme4” package (version 1.1–34; Bates et al., 2015), which has been widely employed in experience sampling studies (Johannes et al., 2021; Smith et al., 2018). All graphs were created using the “sjPlot” package (version 2.8.14) and the “ggplot2” package (version 3.4.0).

Results

As summarized in Table 1, the within-participant correlations indicate that intrinsic motivation ($r=0.50$, $p<0.001$) and identified motivation ($r=0.16$, $p<0.001$) were positively associated with happiness. There was a small positive correlation between external motivation and happiness ($r=0.05$, $p<0.05$). Amotivation was negatively correlated with happiness ($r=-0.19$, $p<0.001$).

We conducted multilevel regression and found that momentary focus positively predicted happiness (estimate = 0.25, SE = 0.03, $t=6.90$, $p<0.001$; see Fig. 1). This result showed that even though the relationship between momentary focus and happiness varies among people, on average, momentary focus led to a higher level of happiness

Table 1 Within-participant correlation

Variable	<i>M</i>	<i>SD</i>	Correlation					
			1	2	3	4	5	6
1. Happiness	4.12	1.09	1					
2. Momentary focus	3.93	1.29	0.29***	1				
3. Intrinsic motivation	3.99	1.26	0.5***	0.14***	1			
4. Identified motivation	4.61	1.06	0.16***	0.26***	0.2***	1		
5. External motivation	4.67	1.15	0.05*	0.23**	−0.05*	0.64**	1	
6. Amotivation	2.63	1.26	−0.19***	−0.16***	−0.19***	−0.48***	−0.48***	1

The number of participants was 56, and they completed the questionnaire a total of 1999 times. All the above variables were occasion factors centered around the person's mean before conducting a correlation test

* $p < 0.05$. ** $p < 0.01$ *** $p < 0.001$

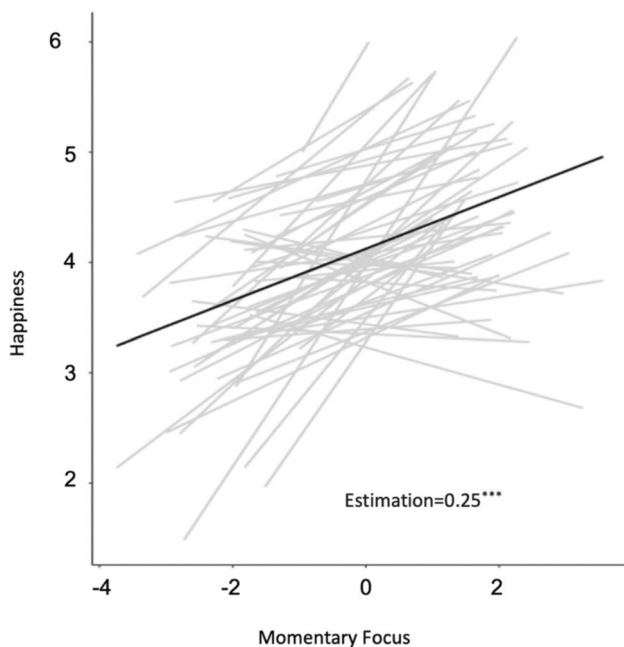


Fig. 1 The estimation shows the main effect of momentary focus on predicting happiness in multilevel regression before controlling the slope and intercept of different participants. The black line represents the estimation of the momentary focus to predict happiness at the within-participant level. The gray line represents the relationship between momentary focus and happiness for 56 participants. *** $p < 0.001$

in daily life. Even after controlling for all motivational factors, momentary focus remained positively associated with happiness (estimate = 0.19, $SE = 0.03$, $t = 6.05$, $p < 0.001$).

Figure 2 displays the scores for momentary focus and happiness across different activities. In this study, the three activities associated with the highest levels of happiness—Conversation, Eating, and Entertainment—were also found to be significantly above average in the research by Killingsworth and Gilbert (2010). Likewise, the activities associated with the lowest levels of happiness in this

study—work-related activities, commuting, and sleeping—were also significantly below average in the Killingsworth and Gilbert (2010) study.

The chart reveals that participants exhibited high momentary focus and happiness during activities such as conversation and entertainment. However, despite individuals reporting a high state of momentary focus during work-related and study activities, they did not experience similarly high levels of happiness. However, even when performing the same task, such as working, people may have different motivations for doing the same thing. For example, some may enjoy learning during class sessions (intrinsic motivation), while others may attend classes merely to complete their academic requirements (external motivation). We further analyzed whether people's different motivations for activities affect the relationship between momentary focus and happiness.

Table 2 shows the moderation effect of motivation on the relationship between momentary focus and happiness. Consistent with self-determination theory, intrinsic motivation positively predicted happiness (estimate = 0.36, $SE = 0.03$, $t = 13.74$, $p < 0.001$). However, amotivation negatively predicted happiness (estimate = -0.07 , $SE = 0.02$, $t = -2.92$, $p = 0.005$). In addition, we found an interaction effect between momentary focus and external motivation (estimate = -0.07 , $SE = 0.03$, $t = -2.77$, $p = 0.01$). However, the interaction term of momentary focus and other motivations did not reach significance ($p > 0.05$). Simple slope analysis revealed that external motivation moderated the relationship between momentary focus and happiness, with higher external motivation weakening this positive association (Fig. 3). No interaction effects were found between momentary focus and other motivational types (intrinsic, identified, and amotivation). These findings indicate that external motivation uniquely attenuates the positive effect of momentary focus on happiness.

Since this study identified the moderating effect of external motivation on the relationship between focus and

Fig. 2 Scores of momentary focus and happiness across different activities. This distribution graph only displays 12 activities with more than 40 data entries to ensure the chart is representative. The horizontal and vertical axes represent the average levels of happiness and momentary focus experienced by all participants during these activities, respectively. The larger the circles, the larger the sample size for each activity

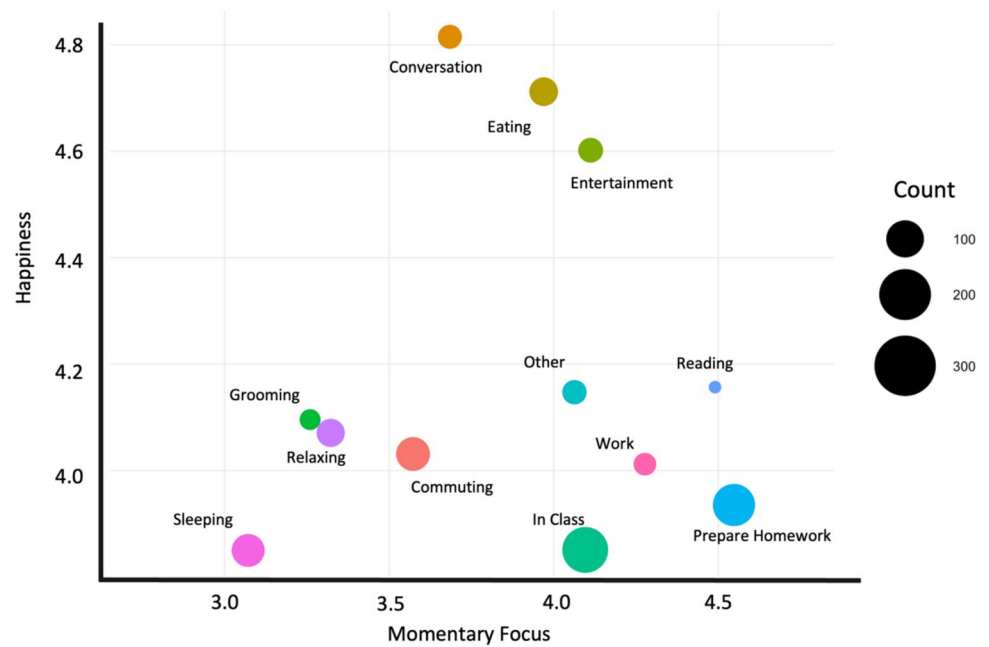


Table 2 Estimation from multilevel model: the main effect and the interaction of momentary focus and motivation to predict happiness

Happiness				
Predictor (fixed effect)	Estimation	SE	<i>t</i>	<i>p</i>
Intercept	4.15	0.08	52.17	<0.001***
Momentary focus	0.20	0.03	6.63	<0.001***
Intrinsic motivation	0.36	0.03	13.74	<0.001***
Identified Motivation	−0.04	0.03	−1.36	0.178
External motivation	−0.01	0.03	−0.17	0.864
Amotivation	−0.07	0.02	−2.92	**
Intrinsic * Focus	0.00	0.02	−0.05	0.961
Identified * Focus	0.02	0.03	0.64	0.525
External * Focus	−0.07	0.03	−2.77	0.010*
Amotivation * Focus	−0.01	0.03	−0.99	0.334
Full model pseudo- <i>R</i> ²	0.336			

The data of 56 participants (35.4 occasions per person) were analyzed. The slope and intercept of all within-participant factors are nested by person level. The random effects of the model have been omitted for clear presentation.

* $p < 0.05$. ** $p < 0.01$. *** $p < 0.001$

happiness, it further examines whether trait mindfulness amplifies this moderating effect. Table 3 shows a three-way interaction between an individual's trait mindfulness, external motivation, and momentary focus to predict happiness (Estimate = -0.08 , $SE = 0.03$, $t = -2.99$, $p = 0.004$). As shown in Fig. 4, external motivation moderated the focus-happiness relationship differently across trait mindfulness levels. For individuals with high trait mindfulness, higher external motivation weakened the positive association

between momentary focus and happiness. In contrast, the focus-happiness relationship remained stable across all levels of external motivation for individuals with low trait mindfulness.

Discussion

We employed experience sampling over a 2-week period to investigate the relationships between momentary focus, happiness, and motivation in daily life. In line with Hypothesis 1, momentary focus positively predicted happiness in participants' daily experiences. This finding aligns with prior research on flow experience and state mindfulness that identifies momentary focus—a core component of both states—as a key enhancer of well-being (Csikszentmihalyi & LeFevre, 1989; Ryan & Deci, 2000; Schutte & Malouff, 2023; Tao et al., 2022). Notably, the positive association between momentary focus and happiness remained significant after controlling for both individual differences and situational motivation.

The results partially support Hypothesis 2, which proposed that external motivation negatively moderates the relationship between momentary focus and happiness. These findings align with previous research in the field. For instance, Schumer et al. (2018) demonstrated that mindfulness training more effectively reduces negative emotions in community populations compared to student populations, attributing this difference to students' tendency to approach mindfulness with external motivations, which may diminish its benefits. This pattern echoes Deci's (1971) seminal work showing that external rewards reduce engagement

Fig. 3 The moderating effect of external motivation on the relationship between momentary focus and happiness. The momentary focus and external motivation were centered by person mean. The simple slope of external motivation at $M - 1SD$ (red line) and $M + 1SD$ (blue line) refers to the value of external motivation below and upon one standard deviation to the mean

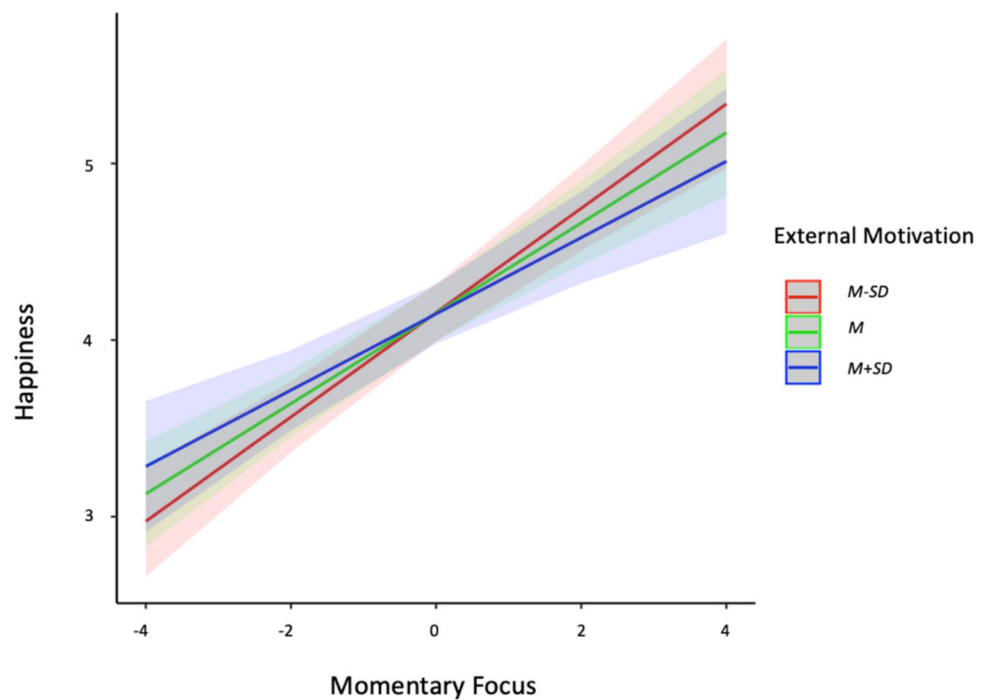


Table 3 Estimation from multilevel model: main effect and the three-way interaction effect of focus, external motivation, and trait mindfulness to predict happiness

Happiness				
Predictor (fixed effect)	Estimation	SE	<i>t</i>	<i>p</i>
Intercept	4.15	0.08	54.69	<0.001***
Momentary focus	0.26	0.04	7.01	<0.001***
External motivation	−0.01	0.04	−0.26	0.800
Mindfulness trait	0.36	0.14	2.52	0.015*
External * Focus	−0.05	0.03	−2.66	0.009**
Mindfulness* Focus	−0.04	0.07	−0.63	0.531
Mindfulness * External	0.00	0.05	−0.02	0.996
Mindfulness * External * Focus	−0.08	0.03	−2.99	0.004**
Full model pseudo- <i>R</i> ²	0.146			

The data of 56 participants, with a mean of 35.4 occasions per person, were analyzed. Momentary focus and external motivation were centered by the person mean, while trait mindfulness was centered by the grand mean. The slope and intercept of all within-person factors were treated as random effects. The random effects of the model were omitted for clear presentation.

* $p < 0.05$. ** $p < 0.01$. *** $p < 0.001$

in inherently interesting activities by undermining intrinsic motivation. Additionally, external motivation creates a reward system focused on performance outcomes, leading individuals to seek happiness from future achievements rather than engaging fully in the present moment (Morgan, 1984; Murayama et al., 2010).

Individuals propelled by external motivations to partake in specific activities may momentarily focus on the immediate task to fulfill their obligations. However, this compelled, momentary focus may not translate into genuine enjoyment, leading to a lack of sustained commitment and an inability to persevere in the activity over the long term. Our findings lead to a deeper reflection on conventional reinforcement learning. Reinforcement learning theory suggests that performance-based rewards can enhance engagement, as evident in students' increased willingness to learn driven by favorable grades (Dayan & Niv, 2008). Paradoxically, our study's findings infer that although such students may transiently intensify their focus on the continuous learning process, their sense of happiness may be diminished.

However, this study did not find intrinsic motivation and identified motivation to enhance the positive link between momentary focus and happiness, not supporting the other part of Hypothesis 2. This result differs from previous research suggesting a positive relationship between autonomous motivation and trait mindfulness (Donald et al., 2020; Li et al., 2023). A possible reason for this discrepancy is that trait mindfulness, compared to momentary focus, involves greater awareness of inner thoughts, emotions, and non-judgmental cognition, which may be more closely related to intrinsic and identified motivation (Ryan et al., 2021). Additionally, this study did not find that amotivation weakens the positive connection between momentary focus and happiness. This could be due to the insufficient representation of participants with high levels of amotivation in the sample,

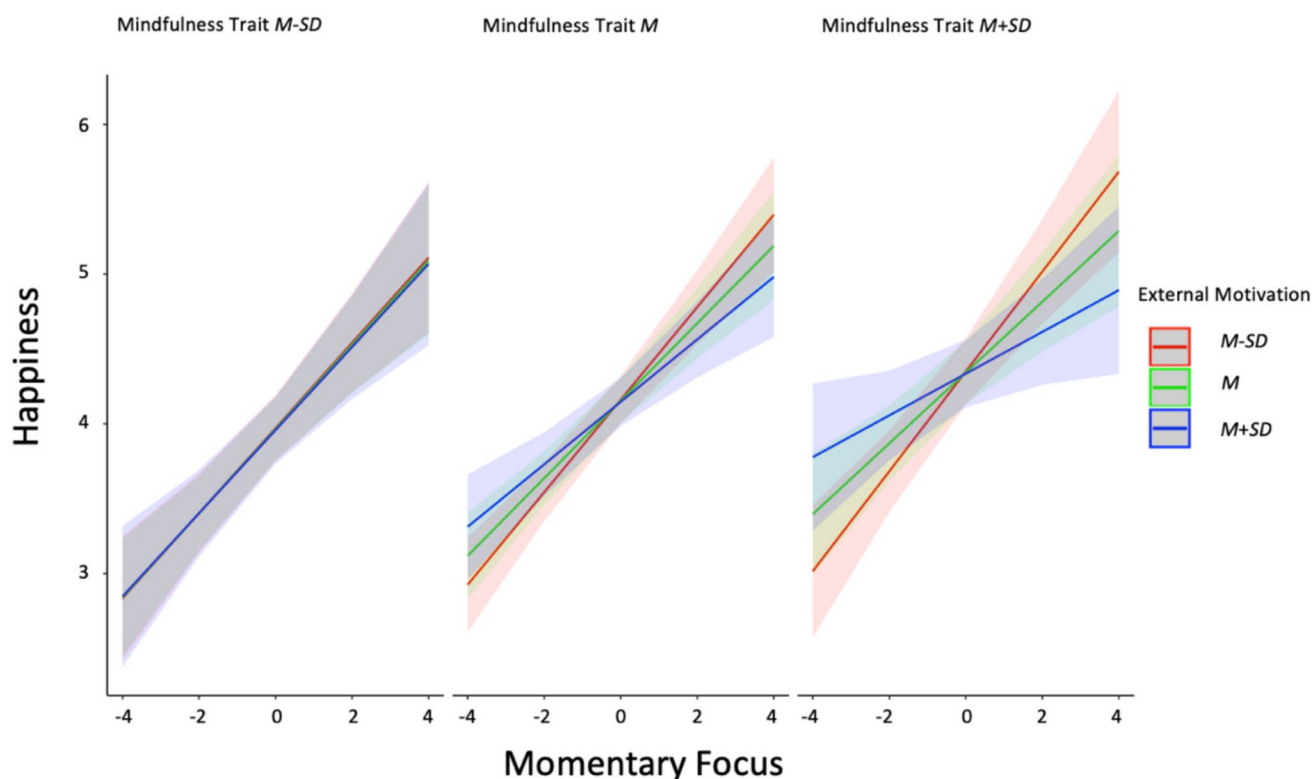


Fig. 4 The three-way interaction of momentary focus, external motivation, and trait mindfulness on predicting happiness. Momentary focus and external motivation were centered by the person mean, while trait mindfulness was centered by the grand mean. The left and right pictures, respectively, show the moderation effect of external motivation for people with trait mindfulness at one standard deviation

below the mean ($M-1SD$) and one standard deviation above the mean ($M+1SD$). The simple slopes of external motivation at $M-1SD$ (red line) and $M+1SD$ (blue line) refer to the values of external motivation below and above one standard deviation from the mean

making it difficult to capture its impact on the relationship between momentary focus and happiness.

Furthermore, the results of the relationship between motivation and happiness generally supported the self-determination theory, which argues that higher inner control of motivation, like intrinsic motivation and identified motivation, brings happiness (Nix et al., 1999; Roussillon Soyer et al., 2022; Yoo, 2015), and lack of motivation decreases happiness (Cui et al., 2024; Legault et al., 2006). These findings suggest that although intrinsic motivation, identified motivation, and amotivation do not influence the relationship between momentary focus and happiness, they can independently impact individuals' happiness. This implies that intrinsic motivation, identified motivation, and amotivation may affect happiness through pathways other than momentary focus.

This study's findings also support Hypothesis 3: Individuals with higher trait mindfulness experience a stronger negative moderating effect of external motivation on the relationship between momentary focus and happiness. This may be due to individuals with a high trait mindfulness possessing

greater metacognitive abilities and having a clearer awareness of their internal thoughts and motivations, thereby enhancing the negative moderating effect of external motivation (Norman, 2017; Ryan et al., 2021).

The results of this study are particularly applicable in the context of mindfulness training. A brief 10-min breath-counting exercise has been shown to promptly enhance the performance of classical attention tasks such as the Filter Task and the Flanker Task (Gorman & Green, 2016). Similarly, just 3-min of mindfulness can significantly increase momentary focus during mundane tasks like dishwashing (Hanley et al., 2015). An individual's ability for attentional control can be further developed through mindfulness interventions like focused attention meditation (Wadlinger & Isaacowitz, 2011). Mindfulness interventions have now been integrated into the education systems of primary and secondary schools in some areas of the USA and Canada, becoming part of the curriculum (Bostic et al., 2015), and it also plays an important role in employee training (Johnson et al., 2020). Therefore, in these settings, it is essential to consider strategies for disentangling people's evaluation

of mindfulness from external incentives, such as academic credit rewards or corporate mandates.

Limitations and Future Research

It is pertinent to acknowledge the limitations of our experiment, and we anticipate that future research will address these gaps, thereby providing deeper insights. Firstly, due to constraints inherent in experience sampling, our study employed a single-item representation for each psychological construct (Bergkvist & Rossiter, 2009). Future endeavors could potentially enhance the reliability and validity of findings by employing multiple-item measures to represent each of these constructs. Secondly, our methodology, which involved participants selecting a dominant activity to assess motivation, did not account for instances of simultaneous engagement in multiple activities (e.g., watching TV while eating). As such, our study did not explore whether the motivation for subordinate activities in multitasking could influence the association between momentary focus and happiness. Future research could investigate the impact of motivation in scenarios involving multitasking. Thirdly, although the momentary focus measured in this study shares many similarities with flow experience, past scholars have suggested that flow experience includes additional dimensions, such as the balance between challenge and skill (Csikszentmihalyi, 1990). Whether the conclusions of this study can be fully generalized to flow experience remains a question for future research to explore.

Fourthly, owing to the limitations of the Situational Motivation Scale, the study did not encompass “introjected motivation,” a distinct form of controlled motivation characterized by internal guilt or pressure (Guay et al., 2000; Ryan & Deci, 2000). Given the potential similarity in impact between introjected motivation and external motivation on happiness (Gillet et al., 2013), future investigations could ascertain whether introjected motivation similarly moderates the relationship between momentary focus and happiness. Finally, this study did not observe a moderating effect of intrinsic motivation, identified motivation, or amotivation on momentary focus and happiness. This may be due to the lack of an intervention-type design in this study. Future research could incorporate an intervention design and enhance these specific types of motivation within certain activities to examine whether relevant moderating effects exist.

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Author Contribution Both authors contributed to the study’s conception and design. Material preparation, data collection, and analysis were performed by Peng-Yu Zeng. The first draft of the manuscript was

written by Peng-Yu Zeng and Su-Ling Yeh. Both authors commented on various versions of the manuscript. Both authors read and approved the final manuscript.

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Data Availability All de-identified raw data and code were stored in the OSF planform (https://osf.io/jyb8f/?view_only=09ad309d75944f74b36bb3a3a5700c10).

Declarations

Ethical Statement This research received ethical approval from the Research Ethics Committee of National Taiwan University (NTU-REC No: 202205HS011).

Informed Consent All participants filled out online informed consent forms before participating in the study.

Conflict of Interest The authors have no conflicts of interest to disclose.

Use of Artificial Intelligence This study used AI tools (ChatGPT) to check English grammar and enhance the readability of the manuscript.

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